WIUT HACKATHON 2026 · COMPUTER VISION FINAL

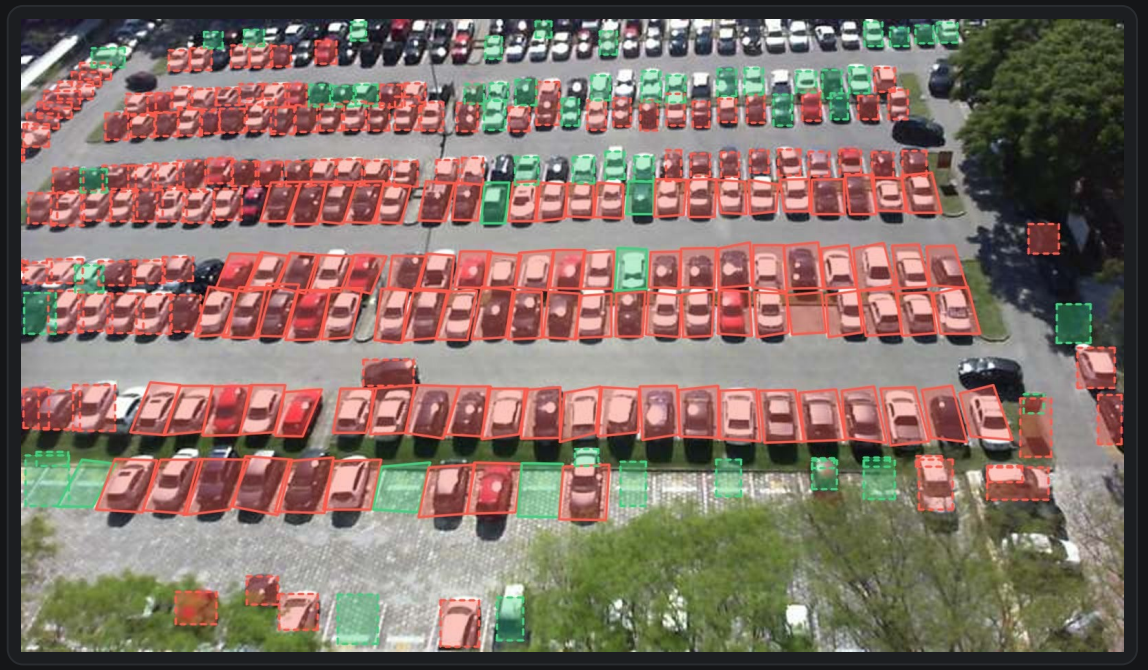
OQIM

A parking tracker from car park cameras

MODE: STARTUP

Team PariVision
Amal Karimov, Komronbek Qodirov, Aziza Adizova
Webster University in Tashkent

Live https://oqim-audit.vercel.app/tracker
Code https://github.com/baytoooo/oqim



The PUCPR car park camera at 09:04 on 17 October 2012 (PKLot dataset). Green: free, red: taken, as our detector read the frame; dashed: spaces the camera found outside the hand labels.

PUCPR | 09:04 | 55 free of 260

PROBLEM AND USER

Drivers circle because they cannot see free spaces

17 h

a year a US driver spends searching for parking (INRIX, 2017)

~5%

of off-street car parks publish live free spaces (Parkopedia)

13,208

paid spaces on 86 streets planned by Poytaxt Parking, Tashkent

SFpark in San Francisco cut cruising for parking by 50% on its pilot blocks. Sources: INRIX, Parkopedia, Spot.uz, ACCESS Magazine

How a driver finds a space today

- Picks a car park by distance or habit, and learns it is full at the gate.
- Drives on to the next one. While searching, the car is part of the traffic.
- A navigator sends every driver to the same nearest car park.

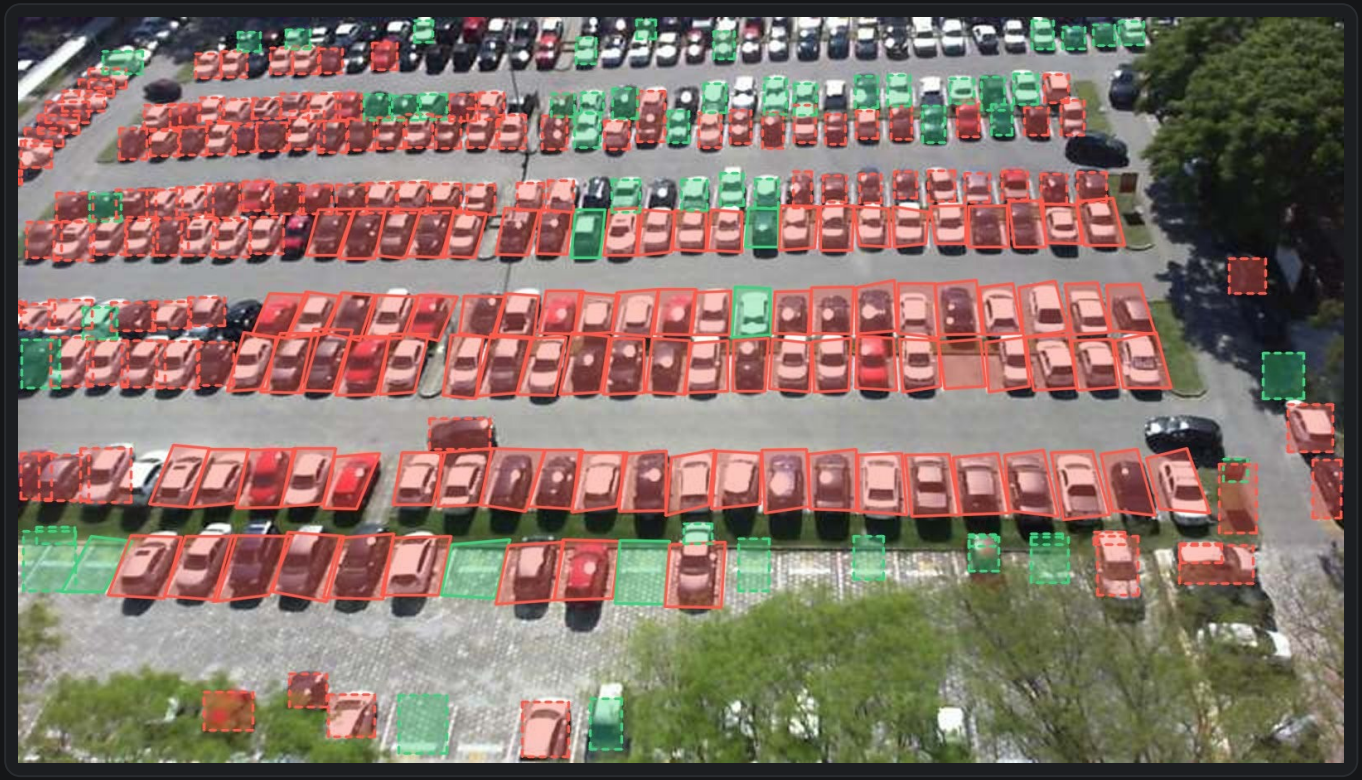
The user

Drivers going to busy places: malls, business centres, hospitals, stations.

Who pays: car park operators, and the city, which wants less searching traffic.

LIVE DEMO · /tracker

Free spaces now, and when you arrive



PUCPR: 100 spaces, 2012-10-17 · UFPR05: 40 spaces, 2013-03-22 · UFPR04: 28 spaces, 2013-01-18

https://oqim-audit.vercel.app/tracker 3 PKLot cameras, 06:55 to 10:35, a frame every 5 min

98.8%

of space readings right: 112 frames, 6,844 readings

1 / 4

free count off by, median / 90th percentile, spaces per frame

3.0 vs 4.6

forecast error 10 min ahead, in spaces; "same as now" gives 4.6

+198 spaces

whole car park: found where cars stay; 146 cars in, 11 out

MEASURED · /parking

97.4% of spaces right on held-out images



Rule: a space is taken when one car, bus or truck box with confidence above 0.05 covers at least 40% of the space's area; the detector runs on the whole frame and on 3 x 3 overlapping tiles.

97.4%

of 13,447 spaces right on 240 held-out images

99.1%

precision, taken

95.3%

recall, taken

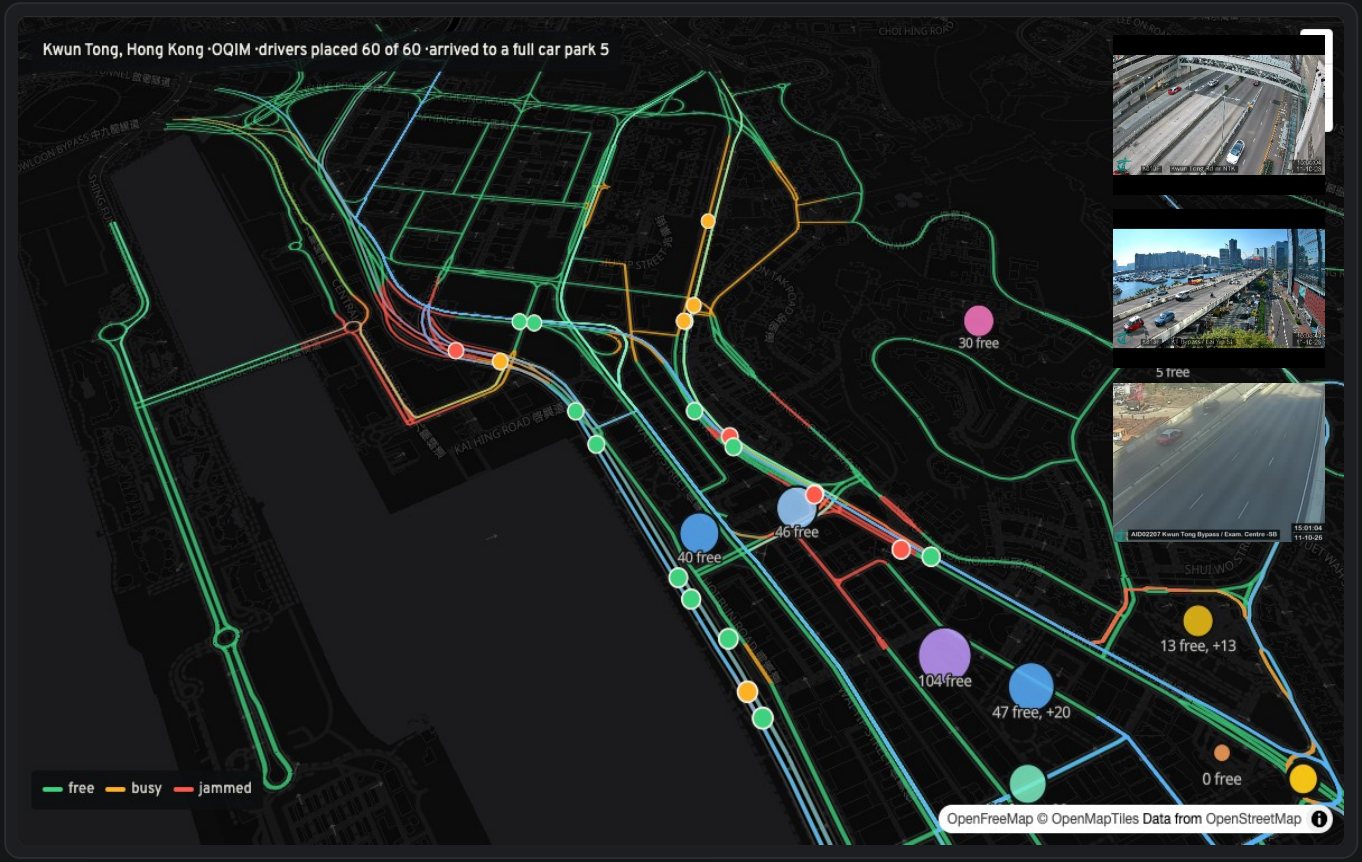
By weather: sunny 97.0%, cloudy 99.0%, rainy 96.1%.

Free count: off by a median of 0, 4 at the 90th percentile.

Not measured: night. PKLot has no night frames.

LIVE DEMO · /city

A district: drivers spread over the car parks



Kwun Tong, Hong Kong: 22 open traffic cameras counted by YOLO26m, live vacancy of 28 car parks (DATA.GOV.HK). Drivers simulated.

60 → 5

drivers arriving to a full car park: nearest car park against OQIM

−18%

door to door, 15.5 to 12.7 min

−41%

vehicle-km with searching, 305 to 181

How: drivers are placed one by one. Each gets the car park and route of least door-to-door time, with the roads and spaces already taken by the drivers placed before.

Snapshot: 13:45 Hong Kong time, 11 October.

From a camera frame to advice

For each car park camera

Frames

One frame every 5 min from an ordinary fixed camera. Labelled spaces, plus the ones found where cars stay 20 minutes.

Detect

YOLO26m on the tracker frames, YOLO26n (10 MB) in the browser; whole frame plus 3 x 3 tiles at 960 px.

Space state

Taken when one box covers 40% of the space. It changes only when two frames in a row agree.

Forecast

Free spaces, cars in and out. Free spaces 10 min ahead from the last 20 minutes' trend and the usual weekday morning.

Advise

Drivers placed one by one on the car park and route of least door-to-door time, counting those already sent.

10 MB

detector, YOLO26n in ONNX Runtime Web: WebGPU or WebAssembly

0

GPU servers. The detector runs in a browser tab.

No plates

and no faces are read. Only boxes over spaces.

Reused, declared: detector and ONNX worker from our elimination system.

Built for the final: tracker, space rule and its test, district model, signal audit.

ALL NUMBERS ARE ASSUMPTIONS TO CHECK WITH THE CUSTOMER

Budget and ROI

Pilot with an operator

3

months, on the operator's own cameras

People: 3 engineers part time. The main cost.

Hosting: static, free to USD 20 a month. No GPU server.

Data: frames from cameras the operator already has. No licences to buy.

Per camera

0

sensors in the ground. One camera covers the spaces it sees.

Our cost: outlining the spaces once, and checking a sample of frames by hand.

Price: per camera per month, set in the pilot.

Value and payback

Value a month =

extra spaces filled × hours × tariff, plus drivers no longer turned away at the gate

Payback = price ÷ value a month

For the city: less searching traffic. The district model counts it in vehicle-km.

We state payback from the pilot inputs, not before.

OUR EDGE

Why it is hard to copy

1

A forecast, not only a count

Free spaces for the minute you arrive, checked against "same as now" (3.0 vs 4.6 spaces off, 10 min ahead).

2

Drivers spread, not herded

Each driver counts against the spaces and roads of the next. A navigator sends all of them to the nearest car park.

3

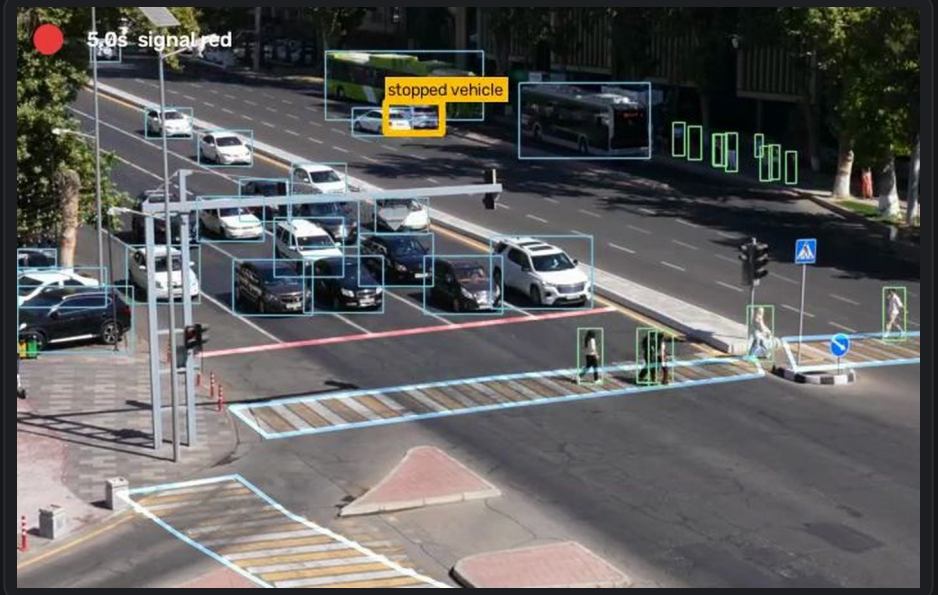
Ordinary cameras

No sensor per space. The detector runs in a browser tab; no plates or faces are read.

4

The same cameras, more products

Our signal audit reads traffic lights from pixels, with no controller access. That is the next stage.



Our lamp reader on the organisers' footage, from pixels only.

98.8%

of frames of C3896: the browser's signal reading agrees with our elimination lamp reader.

NEXT STAGE · /audit

The same cameras can later time the lights

Signal audit from one video: the lights read from pixels, every vehicle over the stop line, then signal plans checked on a replay of the junction's own vehicles. Organisers' footage of one Tashkent junction, southbound.

Clip	Length	Cycle read	Vehicles	Replay check: median departure error	Wait now	Webster (1958)	OQIM predictive
C3896	5:40	75 s	130	**0.50 s**	9.3 s	32.6 s	9.1 s
C3897	5:18	75 s	127	**0.29 s**	10.0 s	29.7 s	8.7 s
C3902	5:18	80 s	170	**0.55 s**	12.3 s	28.8 s	10.1 s
C3905	2:08	80 s	59	**4.54 s**	10.3 s	11.9 s	7.3 s

29.7 vs 10.0 s

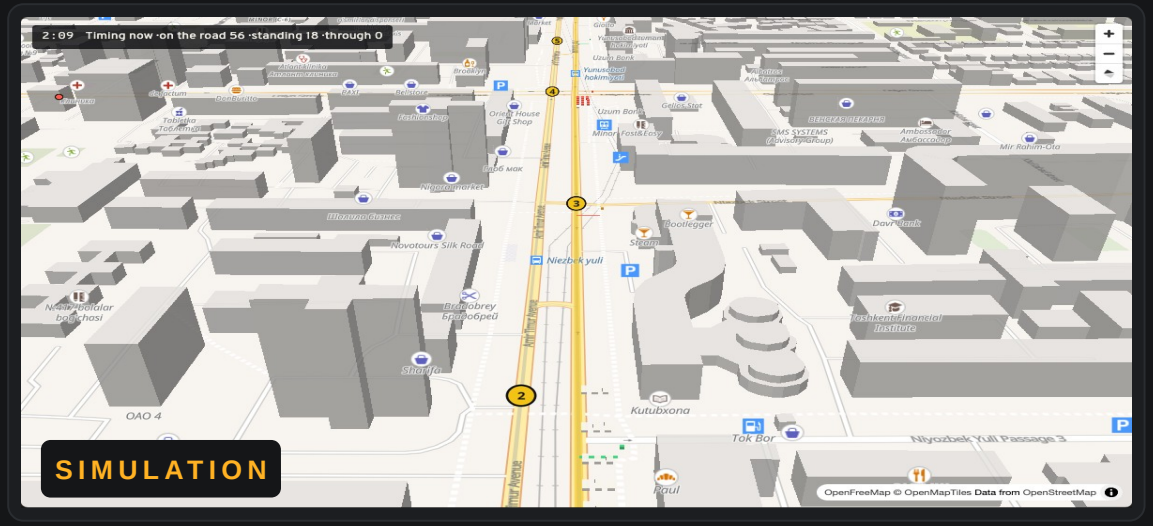
C3897: the textbook plan against the timing now. Worse on all 4 clips.

So OQIM never recommends a plan before it replays it. Mean wait per vehicle on the replay, lower is better. Predictive assumes the next 30 s of arrivals are known, which needs a camera upstream. C3905 (2:08, under 2 cycles) fails the replay check, and the page says so.

Tracks: our elimination pipeline (YOLO26m, 1280 px, 10 fps). Night and rain not yet measured on a test set of our own.

NEXT STAGE · /network AND /roads

A green wave in 3D, and road damage on a map



Amir Temur Avenue, Tashkent: 1.46 km, 5 signals, 224 vehicles in 10 min, Intelligent Driver Model. Green wave against the timing now; only offsets change.



YOLOv8n fine-tuned on RDD2022 (4 classes), in the browser, on phone photos. Measured on a 300-image sample of the RDD2022 test split.

141 s → 24 s
mean delay at the signals

4.58 → 0.60
stops per vehicle

0.66
mAP50

66%
precision

Elsewhere: Copenhagen's green wave cut travel time by 17% (Centre for Public Impact); Google Project Green Light, up to 30% fewer stops (Seattle DOT).

Team PariVision

Amal Karimov
Captain. Scene layout and the signal and vehicle rules

Komronbek Qodirov
Pipeline, live demo and website

Aziza Adizova
Dev set, pedestrian rules and report

1

Pilot

One car park operator, its own cameras, 3 months.

2

Live feed

Frames from a live camera, not a recorded morning.

3

Night test

PKLot has no night frames. A test set of our own, with hand labels.

4

Signals

Then the same cameras time the lights: audit and green wave.

Thank you. Questions?

Try it: https://oqim-audit.vercel.app/tracker Code: https://github.com/baytoooo/oqim